

AI for Interpretable Heart Disease Prediction

Clinical Prediction Report

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Prediction Summary

■ Heart Disease Detected — High Risk

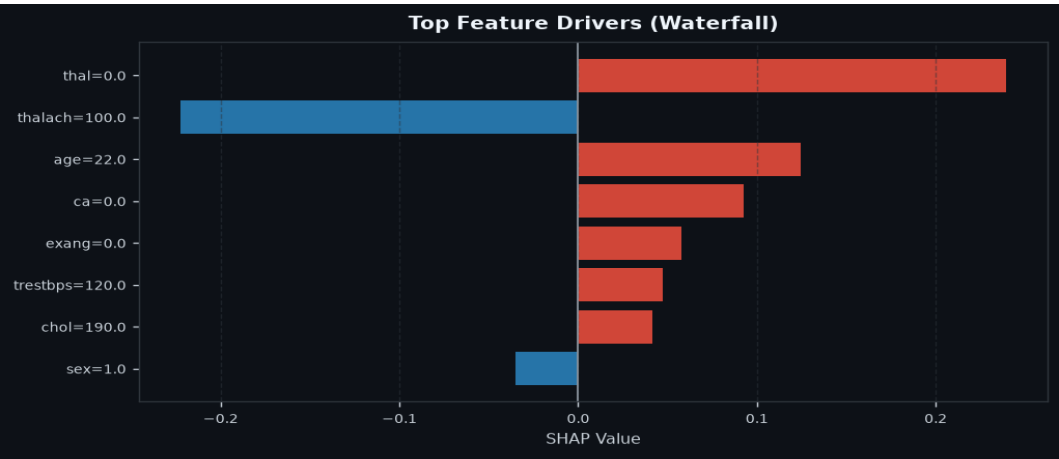
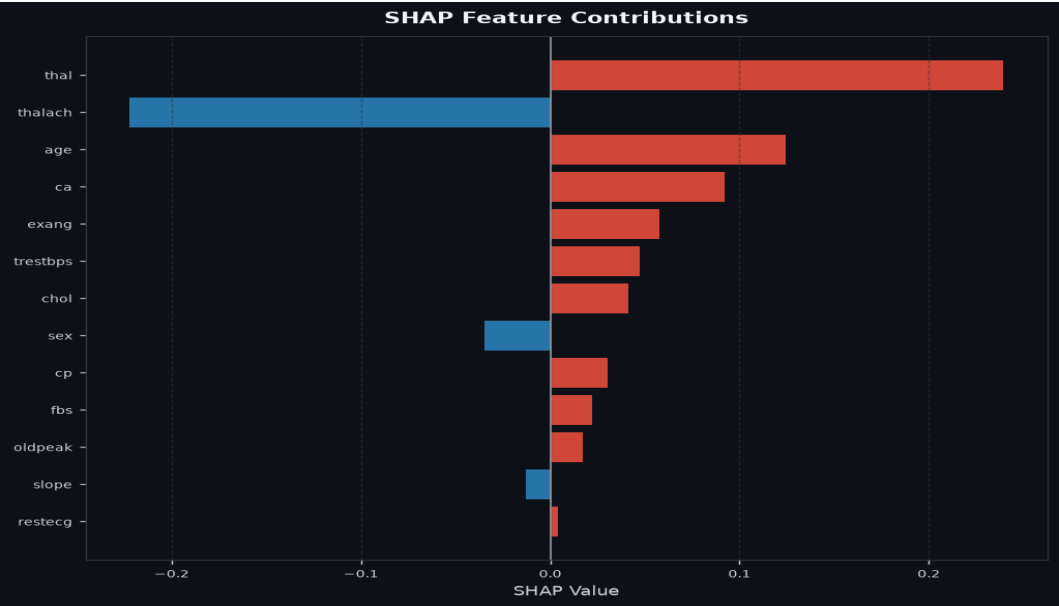
Model	P(Disease)	Weight
Random Forest	78.5%	33%
SVM (RBF)	63.7%	33%
ANN (MLP)	93.9%	33%
Ensemble (Avg)	78.7%	Final

SHAP Clinical Feature Analysis

Feature	Value	SHAP	Direction
Thalassemia (0–3)	0.0	+0.2394	↑ Risk
Max Heart Rate Achieved	100.0	-0.2224	↓ Risk
Age (years)	22.0	+0.1246	↑ Risk
Coronary Arteries Coloured (0–4)	0.0	+0.0924	↑ Risk
Exercise Induced Angina (1=Yes)	0.0	+0.0578	↑ Risk
Resting Blood Pressure (mmHg)	120.0	+0.0472	↑ Risk
Cholesterol (mg/dl)	190.0	+0.0415	↑ Risk
Sex (1=Male, 0=Female)	1.0	-0.0348	↓ Risk
Chest Pain Type (0–3)	1.0	+0.0306	↑ Risk
Fasting Blood Sugar > 120 (1=Yes)	0.0	+0.0221	↑ Risk
ST Depression (oldpeak)	0.0	+0.0171	↑ Risk
Slope of ST Segment (0–2)	1.0	-0.0127	↓ Risk
Resting ECG (0–2)	1.0	+0.0038	↑ Risk

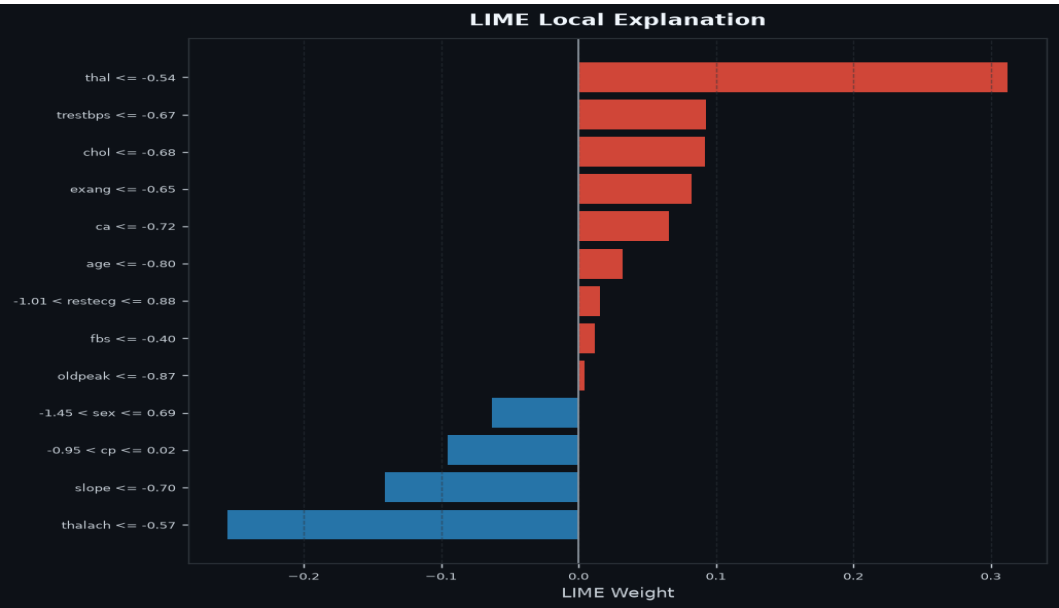
SHAP Explainability

Red bars push prediction toward disease. Blue bars push away from disease.



LIME Local Explanation

Locally fitted linear approximation around this patient's neighbourhood. Positive weights push toward disease.



■ **DISCLAIMER:** This report is generated by an AI research system for educational and research purposes only. It is NOT a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified physician.